Akshay Kumar Mourya

9748925802 akshaymouryaart@gmail.com Kolkata, IN
LinkedIn GitHub

Summary

AI & Machine Learning undergraduate with strong research orientation and hands-on experience in data science, NLP, deep learning, and cloud-based ML deployment. Skilled in building end-to-end ML pipelines and contributing to research-driven development. Seeking entry-level AI/ML or Research roles.

Education

Techno International New Town July 2025
B.Tech in Artificial Intelligence & Machine Learning (CGPA: 8.5 / 10.00) *Kolkata, West Bengal*

Experience

Megaminds IT Service Nov 2025 – Present
Data Science Intern *Remote*

- Conduct applied research in Machine Learning and Natural Language Processing on real-world problem statements.
- Designed and implemented end-to-end ML pipelines including data collection, preprocessing, feature engineering, model training, and evaluation
- Applied NLP techniques such as tokenization, TF-IDF, embeddings, and transformer-based models for text analysis tasks.
- Deployed ML experiments using AWS services (EC2, S3) to enable scalable model testing.

Projects

Movie Recommendation System

- Built a content-based recommender system on a dataset of 5,000 movies using NLP feature engineering.
- Applied **Bag of Words** + **Cosine similarity**, achieving relevant top-5 recommendations.
- Developed an interactive UI using Streamlit for real-time recommendations.
- Used libraries such as NumPy, Pandas, Scikit-Learn, NLTK, and Streamlit to streamline the recommender system's functionality and user interface.

CropInsightX

- Developed a CNN-based plant disease detection system using Python and TensorFlow.
- Designed REST APIs using FastAPI to serve model predictions.
- Integrated blockchain concepts for transparent and secure supply-chain tracking.
- Focused on real-world deployment challenges such as scalability and trust in agrotech systems.

Technical Skills

Programming: Python, C, JavaScript, SQL, HTML/CSS

Machine Learning & DL: Scikit-learn, TensorFlow, PyTorch, CNNs, NLP, Transformers

Data Analysis: Pandas, NumPy, Matplotlib, Seaborn

Web & APIs: Flask, Django, FastAPI, Streamlit

Tools & Cloud: Git, GitHub, AWS (EC2, S3), MySQL

Core Concepts: AI, ML, Neural Networks, OS, Compiler, APIs, Cloud Computing

Achievements & Certifications

- Earned a Badge of Introduction of Data Science - IBM
- Earned a Badge on completion of Introduction of Cloud Computing - IBM

Language

- English [Proficient]

Publications

Empirical Analysis of Heart Disease Detection using Ensemble Learning
*Bikash Sadhukhan, **Akshay Kumar Mourya**, Pratick Gupta, Atulya Narayan, Shivam Kumar*
Published: April 2025
DOI: 10.1142/S0219649225500285